Lymphatic Malformation 13 (LMPHM13): A Comprehensive Disease Characteristics Report

Disease: Lymphatic malformation 13 (LMPHM13) **MONDO ID:** MONDO:0859379 **OMIM phenotype:** #620244 (established; earlier reports used the generic NIHF entry OMIM 236750) **Causal gene: THSD1** (Thrombospondin type-1 domain-containing protein 1), 13q14.3 **Category:** Mendelian, autosomal recessive

Summary

Lymphatic malformation 13 (LMPHM13) is a rare autosomal-recessive lymphatic/vascular disorder caused by biallelic loss-of-function variants in the gene *THSD1* (thrombospondin type-1 domain-containing protein 1), located at chromosome 13q14.3. The gene encodes an 852-amino-acid type-I transmembrane protein that is most highly expressed in lymphatic endothelial cells. LMPHM13 was first linked to THSD1 by Shamseldin and colleagues in 2015 through autozygosity mapping plus exome sequencing in consanguineous families, in which a THSD1 founder mutation accounted for embryonic lethality in three families and a second THSD1 mutation produced a milder phenotype of hydrops fetalis with persistent postnatal edema [PMID: 26036949]. Although "lymphatic malformation 13" is a serial-numbered name that does not itself identify a gene, all identifiers and the literature point consistently to THSD1 — distinct from other numbered lymphatic malformations caused by *PIEZO1*, *FLT4/VEGFR3*, *CCBE1*, *FAT4*, or *EPHB4*.

The clinical picture spans a severity spectrum. At the severe end are embryonic/fetal lethality and non-immune hydrops fetalis (NIHF) with pleural and peritoneal effusions. In survivors, LMPHM13 is a multisystem disorder including recurrent hydrocele, cutaneous/midline hemangiomas, congenital heart defects, intestinal malabsorption/protein-losing enteropathy, and dysmorphic features. At the mild/late end is chronic peripheral lymphedema persisting into adulthood (reported to age 79). Mechanistically, THSD1 stabilizes endothelial focal adhesions by suppressing FAK–Beclin-1–mediated autophagic turnover of those adhesions. When THSD1 is lost, endothelial and lymphatic-endothelial cells detach from the basement

membrane and the vascular/lymphatic barrier fails, allowing fluid and protein to leak into interstitial spaces and body cavities. The same vessel-fragility mechanism underlies an allelic, dominantly-acting association of THSD1 with intracranial aneurysm and subarachnoid hemorrhage, which must be distinguished from the recessive lymphatic phenotype.

Management is entirely supportive — drainage of effusions, albumin replacement, nutritional support, and lymphatic surgery for refractory lymphedema/protein-losing enteropathy. There is no disease-modifying therapy. Prevention rests on genetic counseling with prenatal/carrier testing, particularly in consanguineous families, where most reported cases have occurred. This report synthesizes eight confirmed findings drawn from twelve primary papers covering the disease's genetics, molecular mechanism, cellular origin, clinical spectrum, model organisms, and population genetics.

Key Findings

Finding 1 — LMPHM13 is an autosomal-recessive THSD1-related lymphatic/vascular disorder

LMPHM13 (MONDO:0859379) is caused by **biallelic (homozygous or compound heterozygous) loss-of-function variants in *THSD1***, a gene at chromosome 13q14.3 encoding a thrombospondin type-1 domain-containing transmembrane protein. The gene–disease link was first established by Shamseldin et al. (2015), who used autozygosity mapping combined with exome sequencing in consanguineous families. In their words: *"A founder mutation in one of these genes, THSD1, which has been linked to vascular permeability, accounted for embryonic lethality in three of the study families. Unlike the other six candidate genes, we were able to identify a second mutation in THSD1 in a family with a less severe phenotype consisting of hydrops fetalis and persistent postnatal edema"* [PMID: 26036949]. This single observation demonstrates both the allelic severity spectrum (lethality ↔ survivable edema) and the vascular-permeability basis of the disorder.

Autosomal-recessive inheritance was independently confirmed by Abdelrahman et al. (2018), who reported a novel truncating THSD1 variant that *"cosegregates with the described phenotype in an autosomal recessive mode of inheritance"* in a consanguineous family of four affected individuals [PMID: 30055085]. Multiple subsequent consanguineous-family reports (Al Rawi 2021; Saxena 2023; Lobos Urbina 2025) have reproduced AR inheritance with truncating variants, cementing the gene–disease relationship.

Finding 2 — THSD1 stabilizes endothelial focal adhesions by suppressing FAK–Beclin-1 autophagy

The core cellular function of THSD1 is the stabilization of endothelial focal adhesions (FAs) to the basement membrane. Santiago-Sim et al. (2016), working in an in vitro endothelial model, showed that *"THSD1 loss impaired endothelial cell focal adhesion to the basement membrane. These adhesion defects could be rescued by expression of wild-type THSD1"* but not by mutant THSD1 — directly tying the molecular defect to loss of function [PMID: 27895300].

The downstream molecular logic was defined by Xu et al. (2024): *"THSD1 negatively regulates Beclin 1, a central autophagy regulator, at FAs through interactions with focal adhesion kinase (FAK). THSD1 inactivation diminishes FAK activity and relieves its inhibitory phosphorylation on Beclin 1. This, in turn, promotes the complex formation between Beclin 1 and ATG14"* [PMID: 38396816]. In normal cells, THSD1 keeps FAK active, FAK phosphorylates and inhibits Beclin-1, and autophagic turnover of focal adhesions is suppressed — maintaining stable endothelial attachment and barrier integrity. When THSD1 is lost, FAK activity falls, Beclin-1 is de-repressed, the Beclin-1/ATG14 autophagy-initiation complex forms, and focal adhesions are turned over by autophagy independently of the mTOR/AMPK axis. The 2025 review summarizes this as THSD1 *"promotes focal adhesion assembly and suppresses autophagy-mediated adhesion turnover, thereby stabilizing endothelial attachment and maintaining barrier function"* [PMID: 40564011].

Finding 3 — Clinical spectrum: NIHF with effusions, hydrocele, hemangiomas, cardiac defects, and chylous/protein-losing complications

Across case reports, LMPHM13 presents a recurrent multisystem phenotype. Abdelrahman et al. (2018) provided the canonical description in four affected children: *"Fetal onset pleural and peritoneal effusions were detected in all four patients and were born with moderate to severe hydrops fetalis that resolved with age. Follow up showed relatively normal growth and development apart from mild ascites and haemangiomas in all affected children, recurrent hydrocele in all affected males, intestinal malabsorption in two patients, dysmorphic features in two patients, and congenital cardiac defects in three out of four patients"* [PMID: 30055085].

The spectrum widens at both ends. At the severe/perinatal end, Lobos Urbina (2025) reported a premature infant with a novel truncating variant (c.1009delC:p.Gln337Argfs*73) presenting ascites, pleural effusion, bilateral hydrocele, chylothorax, intestinal perforation, patent ductus arteriosus, capillary hemangiomas, dysmorphism, and midline hemangiomas [PMID: 41842780]. Al Rawi (2021) documented an extremely premature (25-week) infant whose course was

complicated by pulmonary hypertension and chronic lung disease [PMID: 33569873]. At the later/milder end, Mortada (2026) documented chronic lower-limb lymphedema with protein-losing enteropathy (PLE) requiring lymphatic surgery: "the successful surgical management of chronic lower limb lymphedema and PLE in a 7-year-old male with a thrombospondin type 1 domain-containing 1 (THSD1) mutation" *[PMID: 42583604]. Amo (2026) extended the age range dramatically with a 79-year-old woman with lifelong peripheral edema caused by a homozygous splice-site variant [PMID: 42714234]. Importantly, NIHF frequently resolves with age** in survivors, which distinguishes THSD1-NIHF from many other lethal hydrops etiologies [PMID: 37993095].

Finding 4 — Model organisms: zebrafish and mouse LOF recapitulate vascular fragility; THSD1 marks HSCs and endothelial cells

THSD1 loss-of-function has been modeled in both zebrafish and mouse. Santiago-Sim et al. (2016) reported that *"In zebrafish and mice, Thsd1 loss-of-function caused cerebral bleeding (which localized to the subarachnoid space in mice) and increased mortality"* — recapitulating the vascular-fragility phenotype [PMID: 27895300]. Genetically engineered mouse models subsequently established a definitive role for Thsd1 in intracranial aneurysm via Thsd1/FAK signaling (reviewed in Xu 2019) [PMID: 31055715].

The gene was originally identified (as *Tmtsp*) through a Venus knock-in reporter mouse. Takayanagi et al. (2006) *"demonstrated its specific expression in hematopoietic and endothelial cells. In lymphohematopoietic cells, Tmtsp was predominantly expressed in hematopoietic stem and progenitor cells"*, marking definitive hematopoiesis in the yolk sac and AGM region [PMID: 16455951]. This establishes both a validated mouse reporter resource and the endothelial/hematopoietic cell-of-origin.

Finding 5 — THSD1 is tolerant of heterozygous LOF (gnomAD pLI ≈ 0), consistent with recessive inheritance

gnomAD constraint metrics (GRCh38; THSD1 = ENSG00000136114, chr13:52,377,167–52,416,373) show THSD1 is **not haploinsufficient**: pLI = 4.0×10^{-9} (≈0); observed/expected LOF (oe_lof) = 0.735 (90% CI 0.559–0.979); LOF-Z = 1.53; missense-Z = 0.46. A pLI near zero means heterozygous carriers tolerate loss of one allele, which is exactly consistent with the recessive inheritance of LMPHM13 — disease requires loss of **both** alleles. Human identifiers confirmed: NCBI Gene **55901**, **HGNC:17754**, OMIM gene **616821**, UniProt **Q9NS62**, cytoband 13q14.3; mouse ortholog *Thsd1* = NCBI Gene 56229, MGI:1929096.

Finding 6 — Protein architecture: pathogenic truncating variants remove the TSP type-1 domain, transmembrane anchor, and cytoplasmic tail

UniProt Q9NS62 defines THSD1 as an **852-aa type-I transmembrane protein**: signal peptide (1–24); extracellular domain (25–413) containing a single thrombospondin type-1 (TSP) domain at residues 340–393; transmembrane helix (414–434); and a large cytoplasmic domain (435–852), with 7 extracellular N-glycosylation sites. Reported pathogenic alleles truncate at or before these features and are predicted to abolish the TSP domain, membrane anchoring, and the cytoplasmic signaling region:

Variant (protein / nucleotide)	Location relative to domains	Predicted effect
p.Glu298Ter (c.892G>T)	Before TSP domain	Loss of TSP, TM, cytoplasmic tail
p.Gln337Argfs*73 (c.1009delC)	At TSP domain start	Frameshift; loss of TSP + downstream
p.Arg388Glnfs*66	Within TSP domain	Truncates TSP + downstream
c.58+2T>G	N-terminal splice site	Early truncation
c.1494del	Cytoplasmic region	Frameshift (ClinVar likely pathogenic)

ClinVar for THSD1 (~60 records) is dominated by VUS (36) and likely-benign (12) entries, with one likely-pathogenic truncating variant (c.1494del) and two "pathogenic" large 13q copy-number variants. Most LMPHM13 family alleles are **private** (family-specific) and absent or under-represented in ClinVar — a recurring theme in ultra-rare recessive disease.

Finding 7 — Single-cell expression is most enriched in lymphatic endothelial cells — a cell-of-origin link to the lymphatic phenotype

Human Protein Atlas single-cell RNA data classify THSD1 as "cell type enhanced," with the **highest expression in lymphatic endothelial cells (74.7 nCPM)**, followed by retinal ganglion cells (46.4), vascular endothelial cells (43.3), astrocytes (42.0), epididymal basal cells (39.6), fibro-adipogenic progenitors (30.7), retinal bipolar cells (27.5), and myosatellite cells (21.7). HPA RNA tissue specificity is "low tissue specificity"; subcellular location is cytosol; GTEx shows broad expression with no single dominant tissue. The lymphatic-endothelial enrichment

provides a direct molecular rationale for why loss of THSD1 produces a lymphatic-dominant clinical phenotype (lymphedema, chylothorax, protein-losing enteropathy), complementing the original endothelial/HSPC expression described by Takayanagi 2006 [PMID: 16455951].

Finding 8 — Identifier correction: OMIM phenotype #620244 does exist

The research template listed OMIM as "not recorded," but MONDO:0859379 ("lymphatic malformation 13") cross-references (retrieved via EBI OLS4) show a **dedicated OMIM phenotype entry #620244** (exact match), plus GARD:0026725, MedGen MEDGEN:1840915, and UMLS:C5830279. Earlier case reports instead cited the older generic NIHF phenotype series OMIM 236750 (used, e.g., by Al Rawi 2021) before the specific LMPHM13 entry was established. The corrected identifier set is given in the Disease Information section below.

Report by Template Section

1. Disease Information

LMPHM13 is a rare, Mendelian, autosomal-recessive lymphatic/vascular dysplasia. It presents prenatally most often as non-immune hydrops fetalis (serous fluid accumulation in ≥2 body cavities/interstitial spaces without red-cell alloimmunization) and, in survivors, as a multisystem disorder with persistent edema, effusions, hydrocele, hemangiomas, congenital heart defects, and protein-losing enteropathy. Information is derived from **individual patient case reports and small consanguineous-family studies** aggregated into disease-level ontology entries (MONDO, OMIM), not from population EHR cohorts.

Key identifiers: - **MONDO:** 0859379 - **OMIM phenotype:** #620244 (corrected; earlier literature used generic NIHF OMIM 236750) - **OMIM gene:** 616821 (THSD1) - **GARD:** 0026725 - **MedGen:** 1840915 / **UMLS:** C5830279 - **HGNC:** 17754; **NCBI Gene:** 55901; **UniProt:** Q9NS62 - **ICD-10/ICD-11, MeSH, Orphanet:** No dedicated LMPHM13-specific code identified; coded under non-immune hydrops fetalis / generalized lymphatic dysplasia.

Synonyms: LMPHM13; THSD1-related generalized lymphatic dysplasia; THSD1-related non-immune hydrops fetalis.

2. Etiology

Causal factor: Purely genetic — biallelic loss-of-function variants in THSD1 [PMID: 26036949]. No environmental, infectious, or toxic cause is implicated.

Genetic risk factors: The principal (and sufficient) risk factor is inheritance of two LOF THSD1 alleles. **Consanguinity** is a major contributing factor — nearly all reported families are consanguineous, producing homozygous truncating alleles via identity-by-descent. A **founder mutation** in THSD1 accounted for lethality across three families in the original study [PMID: 26036949].

Environmental risk / protective factors: None established. No lifestyle, dietary, occupational, or toxic modifiers have been described. **Gene–environment interactions:** none documented; this is a monogenic disorder.

Protective factors (genetic): Because THSD1 is LOF-tolerant in the heterozygous state (gnomAD pLI ≈ 0, Finding 5), heterozygous carriers are unaffected — a single functional allele is protective.

3. Phenotypes

Phenotype	Type	Onset	Frequency	Suggested HPO
Non-immune hydrops fetalis	Clinical sign	Fetal/neonatal	Core feature	HP:0001790
Pleural effusion	Clinical sign	Fetal onset	4/4 (Abdelrahman)	HP:0002202
Ascites / peritoneal effusion	Clinical sign	Fetal onset	High	HP:0001541
Hydrocele (males)	Physical	Congenital	All affected males	HP:0000034
Hemangioma (cutaneous/ midline)	Physical	Congenital	High	HP:0001028
Congenital heart defect	Physical	Congenital	3/4 (Abdelrahman)	HP:0001627
Intestinal malabsorption / PLE	Lab/ clinical	Infancy–childhood	Variable	HP:0002024 / HP:0002243
Peripheral lymphedema	Clinical sign	Congenital–adult	Variable, persistent	HP:0001004
Chylothorax	Clinical sign	Neonatal	Case reports	HP:0010310
Dysmorphic features	Physical	Congenital	Variable	HP:0001999
Prematurity / pulmonary hypertension	Clinical	Perinatal	Case reports	HP:0001662 / HP:0002092

Severity/progression: Highly variable — from embryonic lethality to lifelong mild edema. In many survivors hydrops/effusions **resolve with age** and growth/development are relatively normal [PMID: 30055085, PMID: 37993095]. Lymphedema/PLE can be chronic and progressive, requiring surgery [PMID: 42583604].

Quality-of-life impact: Not formally measured with EQ-5D/SF-36. Chronic lymphedema and PLE impose substantial burden (nutritional compromise, hypoalbuminemia, repeated interventions); perinatal cases carry mortality risk; late-presenting cases may have lifelong unexplained edema [PMID: 42714234].

4. Genetic / Molecular Information

Causal gene: THSD1 (HGNC:17754, NCBI 55901, OMIM gene 616821), 13q14.3. **Variant types:** Predominantly truncating — nonsense, frameshift, and splice-site (see Finding 6 table). **Classification:** Family alleles are mostly private; per ACMG criteria the recurrent truncating alleles are pathogenic/likely pathogenic by PVS1 (LOF is the established mechanism). ClinVar is dominated by VUS. **Allele frequency:** Pathogenic alleles are absent/ultra-rare in gnomAD. **Origin:** Germline. **Functional consequence:** Loss of function (THSD1 is LOF-tolerant in heterozygotes; disease requires biallelic loss). **Modifier genes / epigenetics:** None established. **Chromosomal abnormalities:** Two large pathogenic 13q CNVs are recorded in ClinVar but these are not the canonical LMPHM13 mechanism.

5. Environmental Information

No environmental, lifestyle, or infectious contributors are known. NIHF from THSD1 is monogenic; TORCH infection and chromosomal/metabolic causes must be **excluded** during diagnosis but are not causal to LMPHM13 itself [PMID: 41842780].

6. Mechanism / Pathophysiology

Ordered causal chain:

1. Biallelic LOF mutation in **THSD1** → **leads to** absence/truncation of functional THSD1 protein (loss of the TSP type-1 domain, transmembrane anchor, and cytoplasmic tail). *(Demonstrated — genetics.)*
2. Loss of THSD1 at focal adhesions → **results in** diminished FAK activity. *(Demonstrated in vitro.)*
3. Reduced FAK activity → **relieves** FAK's inhibitory phosphorylation of Beclin-1. *(Demonstrated — Xu 2024.)*
4. De-repressed Beclin-1 → **promotes** formation of the Beclin-1/ATG14 autophagy-initiation complex (mTOR/AMPK-independent). *(Demonstrated.)*

5. Autophagy → **drives** turnover/disassembly of focal adhesions → **impairs** endothelial cell attachment to the basement membrane. *(Demonstrated; rescued by WT but not mutant THSD1.)*
6. Weakened endothelial/lymphatic-endothelial adhesion → **causes** vascular and lymphatic barrier failure (increased permeability, vessel fragility). *(Inferred from cellular defect + vascular-permeability link.)*
7. Barrier failure branches:
8. **Branch A (vascular/lymphatic leak):** → fluid and protein extravasation → **produces** effusions, hydrops fetalis, lymphedema, chylothorax, protein-losing enteropathy. *(Clinical correlation.)*
9. **Branch B (vessel fragility):** → **predisposes** to intracranial aneurysm/subarachnoid hemorrhage (allelic, dominantly-acting trait; modeled in mouse/zebrafish). *(Demonstrated in models.)*

Molecular pathways: FAK (PTK2) signaling; Beclin-1/ATG14 autophagy initiation; focal-adhesion assembly. Related vascular pathways implicated in the aneurysm literature include TGF-β and mechanotransduction [PMID: 31055715]. **Cellular processes:** Autophagy (GO:0006914), focal adhesion assembly (GO:0048041), cell-matrix adhesion (GO:0007160), regulation of vascular permeability (GO:0043114), lymphangiogenesis (GO:0001946). **Cell types:** Lymphatic endothelial cell (CL:0002138), vascular endothelial cell (CL:0000115), hematopoietic stem cell (CL:0000037). **Subcellular:** Focal adhesion (GO:0005925); cytosol (GO:0005829); plasma membrane (THSD1 is a type-I transmembrane protein). **Protein dysfunction:** Loss of function via truncation of the TSP type-1 domain and cytoplasmic signaling region.

7. Anatomical Structures Affected

- **Primary organs/systems:** Lymphatic system (UBERON:0006558 lymphatic part of vasculature) and cardiovascular/vascular endothelium; serous cavities — pleural (UBERON:0002402), peritoneal (UBERON:0002358).
- **Secondary involvement:** Heart (congenital defects; UBERON:0000948), intestine (malabsorption/PLE; UBERON:0000160), scrotum/tunica vaginalis (hydrocele; UBERON:0000203), skin (hemangiomas; UBERON:0002097), lungs (pulmonary hypertension/chronic lung disease; UBERON:0002048), brain vasculature (aneurysm/subarachnoid space; UBERON:0002361).
- **Tissue/cell level:** Endothelial tissue; specifically lymphatic endothelial cells (CL:0002138) and vascular endothelial cells (CL:0000115).

- **Lateralization:** Effusions/edema are typically **bilateral/generalized**; hydrocele may be bilateral.

8. Temporal Development

- **Onset:** Congenital — most often prenatal (hydrops detected on fetal ultrasound, e.g., at 19 weeks' gestation) [PMID: 41842780]. Rare late/adult presentation of chronic edema [PMID: 42714234].
- **Onset pattern:** Insidious/chronic antenatal fluid accumulation; acute perinatal decompensation possible.
- **Progression:** Variable. Severe end = embryonic lethality/stillbirth. Survivors frequently show **spontaneous resolution** of hydrops/effusions with age and relatively normal development [PMID: 30055085, PMID: 37993095]. Lymphedema/PLE may be chronic and lifelong.
- **Critical period:** The fetal/perinatal window is the key period of vulnerability and the main opportunity for intervention (effusion drainage, respiratory support).

9. Inheritance and Population

- **Inheritance:** Autosomal recessive [PMID: 30055085].
- **Penetrance/expressivity:** Highly **variable expressivity** (lethal ↔ mild lifelong edema), even among affected individuals; penetrance of the biallelic genotype appears high but severity is variable.
- **Consanguinity:** Central — most families are consanguineous; founder effects documented [PMID: 26036949].
- **Carrier frequency:** Not formally established; pathogenic alleles are ultra-rare in gnomAD. Heterozygous carriers are asymptomatic (pLI ≈ 0).
- **Epidemiology:** No prevalence/incidence estimates exist — the disorder is ultra-rare with only a few dozen reported cases worldwide.
- **Sex ratio:** No sex bias in disease (autosomal); hydrocele is, by anatomy, limited to males.
- **Geographic distribution:** Reported families are concentrated in populations with high consanguinity (Middle East/North Africa) and scattered individual reports elsewhere (e.g., Latin America; the gene was discovered in Japan-origin reporter work).

10. Diagnostics

- **Prenatal imaging:** Fetal ultrasound detects hydrops (effusions in ≥2 compartments), ascites, pleural effusion, hydrocele, edema.
- **Laboratory:** Hypoalbuminemia (from PLE/leak); exclude anemia/alloimmunization (to establish *non-immune* hydrops); negative TORCH study; normal karyotype/microarray to exclude chromosomal/aneuploidy causes [PMID: 41842780].
- **Genetic testing (confirmatory): Whole-exome or whole-genome sequencing** is the diagnostic modality of choice, having identified THSD1 variants in essentially all reported cases; WGS detected a deep splice variant (c.58+2T>G) in the adult case [PMID: 42714234]. Single-gene THSD1 testing or NIHF/lymphedema gene panels are appropriate once the gene is suspected. Chromosomal microarray/karyotype serve to exclude differential diagnoses.
- **Differential diagnosis:** Other causes of NIHF — cardiac, chromosomal (e.g., Turner syndrome), metabolic/lysosomal storage, other lymphatic dysplasias (e.g., RASopathy-related, PIEZO1/CCBE1/FAT4 generalized lymphatic dysplasias), infectious (TORCH), and alloimmune hydrops [PMID: 37993095].
- **Screening:** Cascade/carrier testing within affected families; prenatal testing in subsequent pregnancies.

11. Outcome / Prognosis

- **Survival/mortality:** Spans embryonic lethality to normal survival. Several reports emphasize a **favorable prognosis** in survivors once hydrops resolves [PMID: 30055085, PMID: 41842780]. Prematurity complications (pulmonary hypertension, chronic lung disease) worsen perinatal prognosis [PMID: 33569873].
- **Morbidity:** Chronic lymphedema, PLE with hypoalbuminemia/malnutrition, recurrent effusions, and surgical needs [PMID: 42583604].
- **Prognostic factors:** Gestational age/prematurity, severity and persistence of effusions, presence of congenital heart disease, and resolution vs persistence of fluid accumulation. No molecular prognostic biomarkers are validated (though more N-terminal/complete truncations may correlate with greater severity — inferred, not proven).

12. Treatment

No disease-modifying therapy exists. Management is supportive and symptom-directed: - **Perinatal/supportive:** Drainage of pleural/peritoneal effusions (thoracocentesis/paracentesis), respiratory support, management of pulmonary hypertension, albumin replacement, and

nutritional support (NCIT: Supportive Care). - **Nutritional:** For PLE — high-protein, medium-chain-triglyceride (MCT) diets to reduce lymphatic load (standard lymphatic-disorder management). - **Surgical/interventional:** Lymphatic microsurgery — **peripheral lymphatic shunts/lymphovenous anastomosis** successfully managed chronic lower-limb lymphedema with PLE in a child [PMID: 42583604]; hydrocele repair as needed; cardiac surgery for structural defects (NCIT: Surgical Procedure). - **Pharmacogenomics / targeted / gene / cell / RNA therapies:** None reported. The FAK–Beclin-1 autophagy axis is a plausible **future** pharmacological target (e.g., autophagy modulation to stabilize focal adhesions) but is purely hypothetical at present.

13. Prevention

- **Primary prevention:** Not applicable beyond reproductive counseling — the disease is congenital and genetic.
- **Genetic counseling:** Central. Autosomal-recessive 25% recurrence risk for carrier couples; particularly relevant in consanguineous families. Carrier/cascade testing and prenatal diagnosis (or preimplantation genetic testing) are available once the familial variant is known.
- **Secondary prevention:** Prenatal ultrasound surveillance in at-risk pregnancies for early detection/management of hydrops.
- **Tertiary prevention:** Nutritional and surgical management to prevent PLE/lymphedema complications.

14. Other Species / Natural Disease

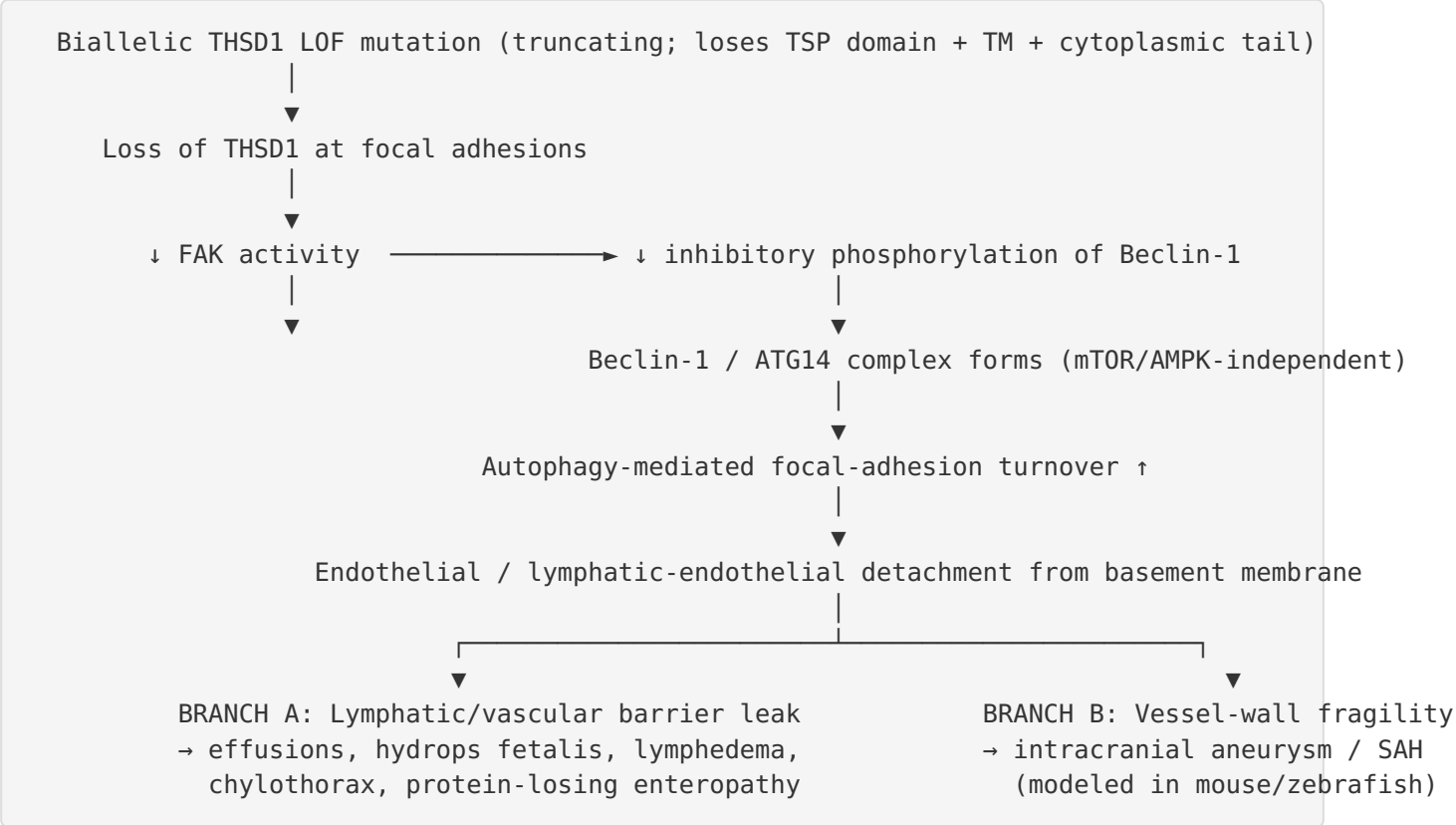
- **Orthologs:** Mouse *Thsd1* (NCBI Gene 56229, MGI:1929096); zebrafish *thsd1* ortholog used in LOF studies. Human NCBI Gene 55901.
- **Natural disease in animals:** No naturally occurring THSD1 disease in companion animals/wildlife is documented in the reviewed literature (no OMIA entry identified here).
- **Comparative biology:** The vascular-fragility phenotype (cerebral/subarachnoid bleeding, increased mortality) is **evolutionarily conserved** across zebrafish and mouse, indicating a conserved role in vascular integrity [PMID: 27895300].
- **Zoonotic potential:** Not applicable (non-infectious genetic disease).

15. Model Organisms

Model	Type	Key phenotype	Reference
Zebrafish *thsd1* LOF	Vertebrate, in vivo	Cerebral bleeding, increased mortality	PMID: 27895300
Mouse *Thsd1* LOF	Mammalian, in vivo	Subarachnoid bleeding; intracranial aneurysm (via Thsd1/FAK)	PMID: 27895300, PMID: 31055715
Tmtsp-Venus knock-in reporter mouse	Mammalian reporter	Marks HSC/progenitor + endothelial cells; definitive hematopoiesis in yolk sac/AGM	PMID: 16455951
In vitro endothelial cells	Cellular	Focal-adhesion defect; rescued by WT THSD1	PMID: 27895300

Phenotype recapitulation: Models reproduce the **vascular-fragility/hemorrhage** axis well (Branch B of the mechanism) but no published model has been reported to fully recapitulate the **hydrops/lymphatic-leak** axis (Branch A) — the dominant human clinical presentation. This is a notable gap. **Resources:** MGI (mouse), ZFIN (zebrafish), Alliance of Genome Resources.

Mechanistic Model / Interpretation



The unifying theme is **loss of endothelial adhesion stability**. THSD1 normally acts as a "brake" on autophagic recycling of focal adhesions via FAK-dependent suppression of Beclin-1. Its loss tips the balance toward adhesion disassembly, destabilizing both lymphatic and vascular endothelium. The single-cell enrichment of THSD1 in **lymphatic endothelial cells** (Finding 7) explains why the lymphatic branch (leak → hydrops/lymphedema/PLE) dominates the human phenotype, while the conserved vascular-fragility branch is most evident in animal models and the allelic aneurysm trait. The severity gradient — from embryonic lethality to lifelong mild edema — likely reflects a combination of allele-specific residual function, modifier background, and stochastic developmental factors, though no modifier has been mapped.

Evidence Base

PMID	Title (abbrev.)	Role in this report
26036949	*Embryonic lethal genes by autozygosity mapping*	**Gene discovery**; THSD1 founder mutation → lethality + milder NIHF/edema; AR
30055085	*Recessive truncating THSD1 variant, 4 patients*	Core phenotype description; AR cosegregation
27895300	*THSD1 mutation in intracranial aneurysm/SAH*	Focal-adhesion defect in vitro; zebrafish/mouse LOF models
38396816	*THSD1 suppresses autophagy via FAK–Beclin-1*	Molecular mechanism (FAK–Beclin-1–ATG14)
40564011	*THSD1 multifaceted regulator (review)*	Synthesis: FA assembly, barrier function, NIHF, tumor suppression
16455951	*Tmtsp marks HSC/endothelial cells*	Cell-of-origin; knock-in reporter mouse
33569873	*THSD1 in extremely premature NIHF infant*	Prematurity, pulmonary hypertension, chronic lung disease
42714234	*Novel THSD1 variant, chronic edema, 79-yo*	Late/adult phenotype extreme; c.58+2T>G splice variant
41842780	*Novel homozygous THSD1 variant, premature NIHF*	c.1009delC:p.Gln337Argfs*73; broad perinatal phenotype
37993095	*Resolving fetal hydrops*	p.Glu298Ter; recurrent NIHF; spontaneous resolution
42583604	*Lymphatic shunts for PLE in THSD1 child*	Chronic lymphedema + PLE; lymphatic surgery
31055715	*Intracranial aneurysms review*	Thsd1/FAK in aneurysm; mouse-model confirmation

No reviewed paper **contradicts** the core model; the literature is consistent and cumulative, moving from gene discovery (2015) → mechanism (2016, 2024) → phenotype expansion and surgical management (2018–2026).

Limitations and Knowledge Gaps

1. **Ultra-rare, no epidemiology:** Only a few dozen cases exist; no prevalence/incidence, penetrance quantification, or genotype–phenotype statistics are possible. All human evidence is from case reports/small consanguineous families — no cohorts or trials.
2. **Lymphatic phenotype not modeled:** Animal models recapitulate vascular hemorrhage/aneurysm (Branch B) but **not** the hydrops/lymphatic-leak phenotype (Branch A) that dominates human disease — the lymphatic branch is inferred from cellular mechanism + expression, not demonstrated in a dedicated lymphatic model.
3. **Variant interpretation:** ClinVar is VUS-dominated; most LMPHM13 alleles are private and absent from population databases, limiting classification support beyond PVS1.
4. **No modifiers/epigenetics:** The basis of the dramatic severity range (lethal ↔ lifelong-mild) is unexplained; no modifier genes, epigenetic marks, or environmental interactions identified.
5. **No QoL or outcome instruments:** No standardized QoL (EQ-5D/SF-36) or natural-history data; survival statistics are anecdotal.
6. **Ontology coding incomplete:** The OMIM gap was corrected (#620244) but no dedicated ICD/Orphanet/MeSH code for LMPHM13 specifically was identified.

Proposed Follow-up Experiments / Actions

1. **Lymphatic-specific model:** Generate a lymphatic-endothelial conditional *Thsd1* knockout (e.g., *Prox1-* or *Lyve1-Cre*) mouse to test whether loss directly causes lymphedema/chylous effusions/PLE — closing the Branch-A gap.
2. **Mechanism in lymphatic endothelium:** Test whether the FAK–Beclin-1–autophagy axis operates in primary human lymphatic endothelial cells, and whether autophagy inhibitors (e.g., FAK-pathway or Beclin-1 modulators) rescue focal-adhesion stability — a potential therapeutic avenue.
3. **Genotype–phenotype registry:** Establish an international THSD1/LMPHM13 registry to aggregate variants, severity, and outcomes; systematically reclassify ClinVar VUS using segregation + functional data.
4. **Functional assay for variant classification:** Develop a standardized focal-adhesion rescue assay (WT vs variant THSD1) to provide PS3-level evidence for VUS reclassification.

5. **Modifier search:** Exome/genome analysis across severity-discordant relatives to identify modifier alleles explaining the lethal-to-mild spectrum.
6. **Natural-history & QoL study:** Prospectively collect longitudinal data (effusion resolution, lymphedema, PLE, growth) with validated QoL instruments to inform counseling and prognosis.
7. **Database curation:** Propagate the corrected identifier (OMIM #620244) and the lymphatic-endothelial cell-of-origin (CL:0002138) into downstream knowledge bases.

Evidence source types: human clinical (case reports/family studies), model organism (mouse/zebrafish), in vitro (endothelial cell assays), and computational (gnomAD constraint, UniProt/HPA annotation, ontology cross-references). All mechanistic and clinical claims are cited to primary literature by PMID.
